

Project Update 3

ECON 345

Due: 11:59PM October 3, 2026

Instructions (Do Not Skip)

Completing a thoughtful research project doesn't happen overnight. Accordingly, we have designed 6 project update assignments that will guide you through the research process. These project updates build upon each other, meaning that you will need to complete Project Update #3 in order to successfully accomplish Project Update #4. Once you submit a project update, your project buddy and economist adviser will provide feedback on your work. If you thoughtfully complete all 6 project updates and routinely incorporate feedback from your buddy and adviser, you will be on track for a solid final research presentation and paper.

The goal of Project Update #3 is for you to begin to understand the relationships between your dependent and independent variables. To that end, you will:

- Compute and interpret conditional summary statistics
- Create and interpret data visualizations

You will not receive credit for work that is missing or incomplete without an explanation of what you tried and how you're struggling. For any parts you are unable to run in RStudio and produce the output for, please include the code (even if it does not work) and briefly explain what you did to attempt the task, and what problems arose (were there specific error messages?). In order to receive partial credit, you must include the code you attempted.

If you run into any difficulties while trying to complete this assignment, consider the following resources:

- Email your adviser (w. econ345.spelman@gmail.com cc'd) with questions on research design. They want to hear from you often!
- For technical help, post a question on Canvas, email econ345.spelman@gmail.com, or come to office hours. Atlanta Fed staff are eager to help!

Instructions for completing this assignment (do not skip)

For this assignment, you will need to submit i) your RMarkdown (RMD) file and ii) the resulting PDF or Word document.

RMD File

The RMD file is where you will compose your project's production code alongside written interpretation in order to easily and clearly communicate your research. As with Project Update #2, organize pieces of your code (from this week's section of your lab notebook i.e., R script) into well-labeled chunks throughout your RMD file, then write your response/interpretation either above or below each chunk. Ideally, you should knit your RMD to PDF so we can easily view your results and provide feedback. Alternatively, you can knit to Word. **If you cannot knit, submit screenshots of your output and your interpretation in a word document.**

Part A - Conditional Summary Statistics (40 points)

1. Compute summary statistics for your dependent variable grouped by the factors of your key independent variable(s). For example, if you're interested in studying the relationship between income and education, you may want to compute median income for different education groups (i.e., people with a college degree vs. without). Contact your adviser if you have questions regarding computing conditional summary statistics for your project. (20 points)
 - If your dependent variable (Y) is continuous and your independent variable (X) is categorical, compute statistics of Y for each level of X . See Lecture 3.
 - If both your X and Y are continuous, define some categorical variable (Z) based on X (e.g., below/above the mean or median) and compute summary statistics of Y for each level of Z . See Lecture 3.
 - If your Y is categorical and X is continuous, define some categorical variable (Z) based on X and compute sample shares of Y for each level of Z . See Lecture 3.
 - A straightforward way to create summary tables in R is by using the `vtable` package, though this is not required — see `summary_tables_with_vtable_package.pdf`, distributed with Lecture 4's materials, for an example.

compute conditional summary statistics

2. Provide an interpretation of the conditional summary statistics (20 points)
 - Explain your reasoning for how you chose to condition your variables.
 - Were you surprised by your results?
 - How do your findings relate to your research question?

Part B - Data Visualizations (50 points)

Data visualizations are a powerful tool for conveying a story. They can also help you understand your data, detect outliers, and observe relationships between your variables. Some commonly used charts include line graphs for time series, histograms for distributions, and scatterplots for exploring relationships between two numerical variables. In this section, you will create a histogram and one other visual of your choice that is best suited for your data. Your adviser can offer suggestions for useful charts.

Produce and Interpret Histograms (25 points)

3. Pick 2 variables in your dataset that are key to your analysis. For each, produce a histogram (if your data are numerical, consider grouping into bins first — for example, if you have numerical income data, create a new variable with 2 bins for low ($< \$50,000$) and high income ($> \$50,000$)). (10 points)

create histograms and plot them

4. Interpret both of your histograms. For example, what do these plots tell you about the location and spread of the distributions? How are those findings different from or similar to your summary statistics? What new patterns emerge? How do these distributions compare to your expectations (prior to looking at the data)? (15 points)
 - Was there anything else you learned from these histograms that wasn't evident from the summary statistics?
 - Was there anything that surprised you?

Visualizing Relationships (25 points)

5. Think of a key relationship you want to explore and how you would visualize it in a chart. Create the chart and interpret it. Choose from the options below: (10 points)

- **Conditional histogram** — for example, if you're interested in the relationship between education and income, compare income distributions for those with vs. without a high school degree. Consider whether it's useful to overlay one histogram over the other.
- **Scatterplot** — for example, if you're interested in the relationship between two numerical variables, such as income and years of education.
- **A visual of your choosing** — line graph, density plot, etc.

create chart and plot it

6. Interpret the results — e.g., what do they tell you about your relationship of interest? (15 points)

Other questions you may want to think about:

- Was there anything else you learned from these visuals that wasn't evident from the summary statistics?
- Was there anything that surprised you?
- If you produced a scatterplot, what does it reveal about the sign and magnitude of the relationship — positive, negative, or none? Strong or weak?
- What patterns emerge, and how do they differ from your previous findings or hypothesis?
- Are there any obvious outliers? If so, explain.
- What are the limitations of this visualization for examining your relationship of interest? What else could you try to address these limitations (mention these ideas in your email to your adviser)?

Part C - Follow-up With Your Adviser (10 points)

Email your adviser with the knitted output (i.e., the PDF or Word document that results from your RMD). Do not send them your R script file or your RMD, but do submit these on Canvas. In your email, give your adviser a brief summary of what you've done this week and what you've learned about your research question, and include any questions or ideas about where to take your analysis next.

- Email your knitted output to your adviser with econ345.spelman@gmail.com cc'd.
- Continue the conversation with your adviser by summarizing your findings and brainstorming next steps in the body of your email.

To receive credit for this, you must cc econ345.spelman@gmail.com.